

Technology Stack

Date	23 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	3 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Responsive, user-friendly interface for entering HDI indicators and viewing results.
2	Backend / Server-Side	Python, Flask	Lightweight web framework for integrating the ML model and processing requests.
3	Database / Data Storage	CSV Dataset, Pandas	Stores HDI dataset efficiently for preprocessing and prediction.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere	Simple and cost-effective deployment of the Flask application.
5	Version Control & CI/CD	Git, GitHub	Version control, collaboration, and project management.
6	Third-Party APIs / Other Tools	Scikit-learn, NumPy, Matplotlib, Seaborn, Anaconda, Jupyter Notebook	Model training, numerical computation, data visualization, and development environment.

